

Leakage Aware Classification of Dermoscopic Skin Lesions

ICS 471 Deep Learning Project Proposal

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Project summary

I will develop a deep learning model that classifies one dermoscopic image into one of seven skin lesion categories. The project uses HAM10000 and compares a small CNN with a pretrained image model. The design controls data leakage by keeping all images of the same lesion in one split.

1. Problem definition

Skin lesions can look similar even when they belong to different diagnostic categories. This project tests whether visual patterns in dermoscopic images can support automated category prediction.

Input: One RGB dermoscopic image resized to 224x224 pixels.

Output: Probabilities for seven lesion categories and the category with the highest probability.

Task type: Single-label multiclass image classification.

Why it matters: Reliable automated screening research may help prioritize cases, but medical use would require independent clinical validation.

2. Dataset

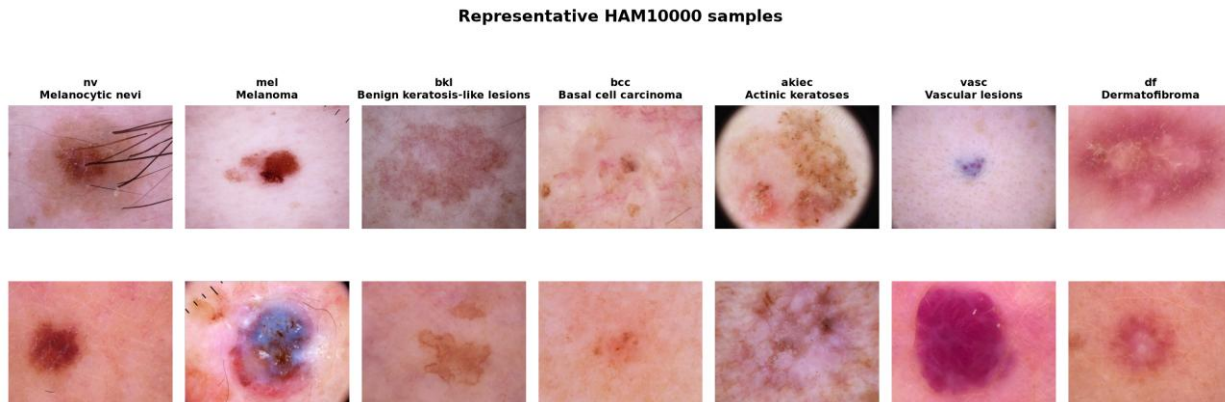
HAM10000 contains 10,015 dermoscopic images collected from two clinical sources. Its metadata includes image ID, lesion ID, diagnosis, confirmation method, age, sex, and anatomical location. More than half of the lesions were confirmed by histopathology; the remaining labels use follow-up, expert consensus, or confocal microscopy.

Access and use: HAM10000 is available through Harvard Dataverse for academic, non-commercial research. Users must accept the dataset terms before downloading. The raw images are not redistributed in this repository.

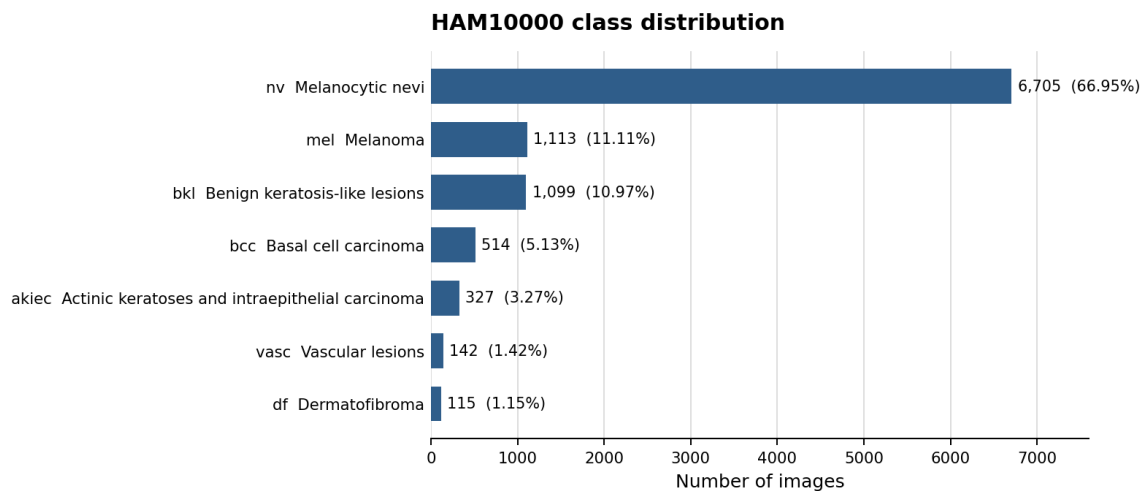
3. Data inspection

The inspection code verifies image availability, reports missing metadata, and creates two proposal figures. Sample inspection should check lesion color, shape, texture, scale, illumination, hair artifacts, and repeated views of the same lesion.

3.1 Representative samples



3.2 Class distribution



Source: HAM10000 metadata. Total = 10,015 images.

The dataset is severely imbalanced. Melanocytic nevi represent 66.95 percent of all images, while dermatofibroma represents 1.15 percent. The largest class is about 58 times the size of the smallest class.

Code	Category	Images	Share
nv	Melanocytic nevi	6,705	66.95%
mel	Melanoma	1,113	11.11%
bkl	Benign keratosis-like lesions	1,099	10.97%
bcc	Basal cell carcinoma	514	5.13%
akiec	Actinic keratoses and intraepithelial carcinoma	327	3.27%
vasc	Vascular lesions	142	1.42%
df	Dermatofibroma	115	1.15%

4. Train validation and test split

The split will be created once before training. Images will be grouped by **lesion_id** so that different views of one lesion cannot appear in different partitions. A stratified group split will preserve the class distribution as closely as the group constraint allows.

Partition	Target share	Use
Training	70%	Parameter learning
Validation	15%	Model selection and early stopping
Test	15%	One final evaluation

- Fixed random seed: 471.
- The split manifests will be saved before model training.
- Normalization statistics, augmentation, class weighting, and oversampling will use training data only.
- The test set will remain untouched until the final model is selected.

5. Validation metric

The single model-selection metric will be macro-averaged F1. An F1 score is calculated for each class, then the seven scores are averaged equally. This prevents the dominant nevus class from controlling the evaluation. The checkpoint with the highest validation macro-F1 will be evaluated once on the test set.

6. Planned modeling

- Baseline: majority-class prediction and a small CNN trained from scratch.
- Main model: transfer learning with ResNet-18 or EfficientNet-B0.
- Training: class-weighted cross-entropy, conservative augmentation, and early stopping.
- Final analysis: test macro-F1, confusion matrix, and per-class error review.

7. Limitations

HAM10000 is curated and does not represent every skin tone, clinic, camera, or population. Some labels are not histopathology-confirmed. Results may not generalize to ordinary phone photographs or clinical deployment. These limits will be stated with the final results.

References

Assistance statement: ChatGPT was used to help organize the proposal and generate initial repository code. I reviewed and validated the code, dataset inspection, figures, and final content.

HAM10000 dataset record. Harvard Dataverse. <https://doi.org/10.7910/DVN/DBW86T>